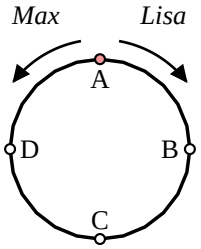
Preview: set-01/by_level/Grade_03-04_Ecolier/y2025_q10 (v1)

Running towards each other around a circle: between two meetings they cover one whole lap together

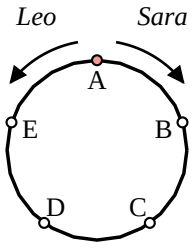
Part A: Each question is worth 3 points

1. Lisa and Max draw a big circle in the sand and write the letters A, B, C and D on it. They start at the same time from point A. Lisa runs clockwise and Max runs anticlockwise around the circle (see picture). They meet for the first time at point B, then at C, at D and then again at point A. How many times has Max run around the circle by then?



A. 1 time ***B.*** 3 times ***C.*** 4 times ***D.*** 5 times ***E.*** 8 times

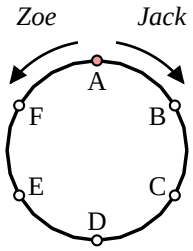
2. Sara and Leo draw a big circle in the sand and write the letters A, B, C, D and E on it. They start at the same time from point A. Sara runs clockwise and Leo runs anticlockwise around the circle (see picture). They meet for the first time at point E, then at D, at C, at B and then again at point A. How many times has Sara run around the circle by then?



A. 1 time ***B.*** 2 times ***C.*** 3 times ***D.*** 4 times ***E.*** 6 times

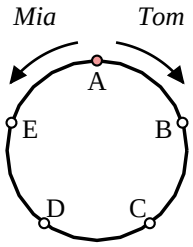
Part B: Each question is worth 4 points

3. Jack and Zoe start at the same time from point A of the circle. Jack runs clockwise, Zoe runs anticlockwise (see picture). They meet for the first time at point E, then at C and then again at point A. How many times has Jack run around the circle by then?



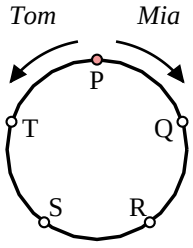
A. 2 times ***B.*** 3 times ***C.*** 4 times ***D.*** 5 times ***E.*** 6 times

4. Tom and Mia start at the same time from point A of the circle. Tom runs clockwise, Mia runs anticlockwise (see picture). They meet for the first time at point D, then at B, at E, at C and then again at point A. How many times has Tom run around the circle by then?



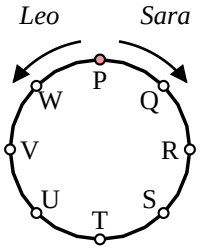
A. 1 time ***B.*** 2 times ***C.*** 3 times ***D.*** 4 times ***E.*** 5 times

5. Mia and Tom start together at point P and run around the circle in opposite directions: Mia clockwise and Tom anticlockwise (see picture). The points are evenly spaced. Mia runs 4 times as fast as Tom. At which point do they meet for the first time?



A. point P ***B.*** point Q ***C.*** point R ***D.*** point S ***E.*** point T

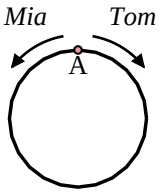
6. Sara and Leo start together at point P and run around the circle in opposite directions: Sara clockwise and Leo anticlockwise (see picture). The points are evenly spaced. Leo runs 7 times as fast as Sara. At which point do they meet for the first time?



A. point P ***B.*** point Q ***C.*** point S ***D.*** point U ***E.*** point W

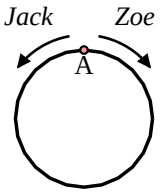
Part C: Each question is worth 5 points

7. Tom runs clockwise and Mia runs anticlockwise around a circle. They start together at point A. When they are together at point A again for the first time, Tom has run around the circle once and Mia 2 times. How many times did they meet? (Count the meeting at A, but not the start.)



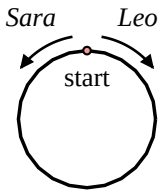
A. 1 ***B.*** 2 ***C.*** 3 ***D.*** 4 ***E.*** 5

8. Zoe runs clockwise and Jack runs anticlockwise around a circle. They start together at point A. When they are together at point A again for the first time, Jack has run around the circle 3 times and Zoe 5 times. How many times did they meet? (Count the meeting at A, but not the start.)



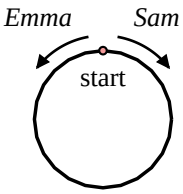
A. 2 ***B.*** 4 ***C.*** 5 ***D.*** 7 ***E.*** 8

9. Leo and Sara start together at the same spot of a round track. Leo runs clockwise and Sara runs anticlockwise. Sara needs 12 minutes to run once around, Leo only needs 4 minutes. After 12 minutes they stop — both are back at the start together. How many times did they meet? (Count the meeting at the end, but not the start.)



A. 1 ***B.*** 2 ***C.*** 3 ***D.*** 4 ***E.*** 5

10. Sam and Emma start together at the same spot of a round track. Sam runs clockwise and Emma runs anticlockwise. Emma needs 8 minutes to run once around, Sam also needs 8 minutes. After 24 minutes they stop — both are back at the start together. How many times did they meet? (Count the meeting at the end, but not the start.)



A. 1 ***B.*** 3 ***C.*** 5 ***D.*** 6 ***E.*** 7

Answer Key

1. **B**	2. **D**	3. **A**	4. **C**	5. **E**	6. **B**
7. **C**	8. **E**	9. **D**	10. **D**		

Solutions

1. B (3 times) — Between two meetings they run exactly once around the circle together. Lisa only gets to the next letter (1 of 4 parts), so Max runs the other 3 parts. 4 meetings × 3 parts = 12 parts = 3 times around.
[set-01/by_level/Grade_03-04_Ecolier/2025 Q10 · fast_laps]

2. D (4 times) — Between two meetings they run exactly once around the circle together. Leo only gets to the next letter (1 of 5 parts), so Sara runs the other 4 parts. 5 meetings × 4 parts = 20 parts = 4 times around.
[set-01/by_level/Grade_03-04_Ecolier/2025 Q10 · fast_laps]

3. A (2 times) — Each time, Zoe runs 2 of the 6 parts and Jack runs the other 4. They meet 3 times, so Jack runs 3 × 4 = 12 parts = 2 times around. [set-01/by_level/Grade_03-04_Ecolier/2025 Q10 · skip_points]

4. C (3 times) — Each time, Mia runs 2 of the 5 parts and Tom runs the other 3. They meet 5 times, so Tom runs 5 × 3 = 15 parts = 3 times around. [set-01/by_level/Grade_03-04_Ecolier/2025 Q10 · skip_points]

5. E (point T) — Until they meet they run once around the circle together, and Mia runs 4 times as far: Tom runs 1 part and Mia 4 parts of 5. 1 part of 5 is 1 of the 5 gaps, so Tom only gets to point T.
[set-01/by_level/Grade_03-04_Ecolier/2025 Q10 · where_first]

6. B (point Q) — Until they meet they run once around the circle together, and Leo runs 7 times as far: Sara runs 1 part and Leo 7 parts of 8. 1 part of 8 is 1 of the 8 gaps, so Sara only gets to point Q.
[set-01/by_level/Grade_03-04_Ecolier/2025 Q10 · where_first]

7. C (3) — Between two meetings they run exactly once around the circle together. Together they ran 1 + 2 = 3 times around, so they met 3 times. [set-01/by_level/Grade_03-04_Ecolier/2025 Q10 · count_meetings]

8. E (8) — Between two meetings they run exactly once around the circle together. Together they ran 3 + 5 = 8 times around, so they met 8 times. [set-01/by_level/Grade_03-04_Ecolier/2025 Q10 · count_meetings]

9. D (4) — Between two meetings they run once around the track together. In 12 minutes Sara runs 1 and Leo runs 3 times around: 1 + 3 = 4 laps together, so 4 meetings. [set-01/by_level/Grade_03-04_Ecolier/2025 Q10 · minutes]

10. D (6) — Between two meetings they run once around the track together. In 24 minutes Emma runs 3 and Sam runs 3 times around: 3 + 3 = 6 laps together, so 6 meetings. [set-01/by_level/Grade_03-04_Ecolier/2025 Q10 · minutes]